

Contents

Chapter 1 – Project Overview & Technical Context.....	3
1.1 Project Goal.....	3
1.2 System Scope	4
Devices Being Monitored	4
Electrical Characteristics.....	4
Deployment Environment	4
1.3 High-Level Technology Stack.....	5
Frontend	5
Backend.....	5
Embedded Firmware	5
Controller Hardware	5
Chapter 2 – Hardware Architecture & Design Approach.....	6
2.1 Hardware Design Philosophy	6
Electrical Safety.....	6
Modular and Replaceable Components	6
Industry-Proven Sensors	6
Readily Available Components	7
2.2 Power Detection Methodology.....	7
Rationale for Current-Based Detection vs Voltage-Based Detection	7
ON / OFF Detection via Current Threshold	7
Event-Driven Reporting (State Change Only).....	8
2.3 Hardware Signal Flow	8
Signal Flow Overview	8
Physical Isolation Between AC Mains and Microcontroller	8
Low-Voltage Operation on Controller Side	9
2.4 Controller Selection Rationale	9
ESP32 Selection Overview.....	9
Wi-Fi Connectivity	9
Arduino Framework Compatibility.....	9
GPIO and ADC Requirements.....	9
Chapter 3 – Hardware Bill of Materials (BOM) & Specifications	10

3.1 Core Control Components	10
Microcontroller Board Requirements.....	10
Power Supply Requirements (AC → DC)	10
3.2 Sensing Components.....	10
Current Sensor Type and Form Factor	10
Measurement Range Requirements	11
Electrical Isolation Requirements	11
3.3 Supporting Components	11
Signal Conditioning Components.....	11
Protection Components	11
Enclosure Requirements	12
Connectors and Wiring	12
3.4 Optional / Alternative Components.....	12
Smart Plug (Commercial Off-the-Shelf Option)	12
Voltage Sensor Alternative (If Required)	12
Industrial-Grade Sensor Options.....	12
3.5 Compliance & Safety Notes	13
Electrical Safety Considerations.....	13
Installation Constraints	13
Environmental Considerations.....	13

Chapter 1 – Project Overview & Technical Context

1.1 Project Goal

The objective of this project is to design and deploy a **reliable IoT-based hardware system** capable of determining the **ON / OFF operational status** of critical electrical devices, specifically:

- Electrical generators
- Uninterruptible Power Supply (UPS) units
- Computer systems and related IT equipment

The system shall detect whether a monitored device is actively powered and report **state transitions (ON → OFF, OFF → ON)** in **real time** to a centralized platform.

In addition to live status visibility, the system will support **long-term operational analytics**, including:

- Automatic calculation of **uptime** (total duration the device is ON)
- Automatic calculation of **downtime** (total duration the device is OFF)
- Timestamped event logging for historical analysis and reporting

From a hardware perspective, the project emphasizes the following design principles:

- **Non-intrusive measurement:**
The monitoring system must not interfere with the normal operation of the electrical devices. Sensors shall be installed externally without modifying the internal circuitry of the monitored equipment.
- **Electrical safety:**
The design must ensure galvanic isolation between high-voltage AC power lines and low-voltage control electronics to minimize risk during installation and operation.
- **Scalability:**
The hardware design shall allow multiple devices to be monitored independently using the same architecture, enabling future expansion without redesign.
- **Reliability and maintainability:**
Components shall be selected based on availability, stability, and suitability for continuous operation in indoor or semi-industrial environments.

The primary function of the hardware subsystem is **state detection and event reporting**, not high-precision energy measurement. This simplifies sensor requirements and improves overall system robustness.

1.2 System Scope

Devices Being Monitored

The system is designed to monitor devices that draw electrical power from standard AC mains, including but not limited to:

- Standalone diesel or petrol generators
- UPS units supplying backup power to IT systems
- Desktop computers, servers, or other critical computing equipment

Each monitored device will be treated as an independent entity, identified uniquely within the system.

Electrical Characteristics

- **Power Type:**
Single-phase AC power
- **Measurement Objective:**
Presence or absence of electrical load (ON / OFF state), rather than exact voltage or power consumption values.
- **Detection Methodology (High-Level):**
ON/OFF status is inferred based on electrical activity (e.g., current flow), allowing safe and indirect monitoring.

Deployment Environment

The system is intended for deployment in:

- Indoor environments (server rooms, offices, control rooms)
- Semi-industrial environments (generator rooms, utility areas)

Environmental assumptions include:

- Moderate temperature and humidity
- No direct exposure to rain or extreme weather
- Availability of standard Wi-Fi connectivity

The hardware shall be enclosed appropriately to protect against dust and accidental contact with electrical components.

1.3 High-Level Technology Stack

While this document focuses on hardware, a high-level overview of the full technology stack is provided to give architectural context.

Frontend

- **React**
- Used to present real-time device status, historical uptime/downtime data, and system dashboards.

Backend

- **Nest.js**
- Responsible for:
 - Real-time data ingestion
 - WebSocket-based updates to the frontend
 - Consumption of device messages via MQTT
 - Storage and processing of device state events

Embedded Firmware

- **Arduino Framework**
- Used to develop firmware for the microcontroller
- Provides rapid development, strong community support, and long-term maintainability

Controller Hardware

- **ESP32 Microcontroller**
- Key characteristics:
 - Integrated Wi-Fi connectivity
 - Sufficient processing capability for sensor data acquisition
 - Multiple ADC and GPIO interfaces
 - Wide industry adoption and strong supply availability

The ESP32 acts as the **edge controller**, interfacing with electrical sensors, determining device state, and transmitting state changes to the backend system over the network.

Chapter 2 – Hardware Architecture & Design Approach

2.1 Hardware Design Philosophy

The hardware architecture for this project is designed to prioritize **safety, reliability, and long-term maintainability**, while remaining flexible enough to support future expansion. The following core principles guide all component and architectural decisions.

Electrical Safety

The system is explicitly designed to **avoid direct exposure to high-voltage AC lines** wherever possible. All sensing and measurement techniques are selected to ensure:

- Electrical isolation between mains voltage and low-voltage electronics
- Reduced risk during installation, maintenance, and operation
- Compliance with standard electrical safety practices

This approach minimizes hazards to personnel and reduces the likelihood of damage to control electronics due to electrical faults or surges.

Modular and Replaceable Components

The hardware architecture follows a **modular design approach**, where each functional block (controller, sensor, power supply) is independent and replaceable. This enables:

- Easy component replacement in the event of failure
- Vendor flexibility during procurement
- Incremental upgrades without full system redesign

Standard interfaces and commonly available modules are preferred over proprietary or tightly integrated solutions.

Industry-Proven Sensors

Only **widely adopted and field-tested sensor technologies** are considered for this system. This ensures:

- Predictable performance in continuous operation
- Availability of technical documentation and community knowledge
- Reduced integration risk

The project avoids experimental or niche sensor solutions in favor of components with established usage in power monitoring and industrial IoT applications.

Readily Available Components

All components are selected based on:

- Availability from multiple suppliers
- Long-term production viability
- Compatibility with global procurement channels

This reduces supply chain risk and ensures consistent replacement availability over the system lifecycle.

2.2 Power Detection Methodology

Rationale for Current-Based Detection vs Voltage-Based Detection

The primary objective of the system is to detect whether a device is **operational (ON)** or **non-operational (OFF)**, not to measure precise electrical parameters.

Two common approaches were evaluated:

Voltage-Based Detection

- Measures the presence of AC voltage
- Requires direct connection to high-voltage lines
- Increases safety risks and isolation complexity
- Voltage may be present even when the device is not actively operating

Current-Based Detection (Selected Approach)

- Detects actual electrical load draw
- Can be implemented using non-invasive sensors
- Provides a more accurate indication of device activity
- Allows physical isolation from mains voltage

Current-based detection was selected as it offers a safer, more reliable indication of device operation while minimizing electrical risk.

ON / OFF Detection via Current Threshold

The system determines device state using a **configurable current threshold**:

- When measured current exceeds the defined threshold → **Device is ON**

- When measured current falls below the threshold → **Device is OFF**

This approach:

- Is computationally simple
- Reduces processing overhead on the controller
- Avoids noise and false positives through threshold tuning

The threshold values can be adjusted during commissioning to match the characteristics of different device types (generator, UPS, computer).

Event-Driven Reporting (State Change Only)

To optimize network usage and system efficiency, the hardware is designed to report **only state changes**, not continuous measurements.

- ON → OFF or OFF → ON transitions generate events
- No periodic data transmission when the state remains unchanged

Benefits include:

- Reduced network traffic
- Lower power consumption
- Cleaner event history for uptime/downtime analysis

2.3 Hardware Signal Flow

Signal Flow Overview

The hardware signal flow follows a simple and robust sequence:

Electrical Device → Sensor → ESP32 Controller → Network

Each stage is logically and electrically isolated to ensure safe and reliable operation.

Physical Isolation Between AC Mains and Microcontroller

The sensing stage provides isolation between:

- High-voltage AC power lines (device side)
- Low-voltage DC electronics (controller side)

This isolation is achieved through:

- Non-invasive current sensing
- Sensor modules with built-in electrical separation

As a result:

- The ESP32 and associated electronics operate exclusively at low voltage
- Electrical faults on the mains side are less likely to propagate to the controller

Low-Voltage Operation on Controller Side

All control electronics operate within standard low-voltage ranges (typically 3.3V to 5V DC). This ensures:

- Compatibility with microcontroller input specifications
- Reduced thermal stress on components
- Simplified power supply design

Low-voltage operation also improves overall system reliability and safety during installation and maintenance.

2.4 Controller Selection Rationale

ESP32 Selection Overview

The ESP32 microcontroller has been selected as the primary controller based on its suitability for network-connected sensing applications.

Key selection criteria include:

Wi-Fi Connectivity

- Integrated Wi-Fi eliminates the need for external communication modules
- Enables direct connection to existing network infrastructure
- Supports real-time data transmission with low latency

Arduino Framework Compatibility

- Mature and widely supported development ecosystem
- Availability of libraries for sensor integration and networking
- Reduces firmware development time and long-term maintenance effort

GPIO and ADC Requirements

The ESP32 provides:

- Sufficient analog-to-digital converter (ADC) channels for sensor input
- Multiple GPIO pins for expansion and auxiliary functions
- Adequate processing capability for signal sampling and state evaluation

Chapter 3 – Hardware Bill of Materials (BOM) & Specifications

3.1 Core Control Components

Microcontroller Board Requirements

The system shall use a **Wi-Fi enabled microcontroller** capable of interfacing with analog sensors and transmitting device state information to a backend system.

Required characteristics:

- ESP32-based development board
- Integrated Wi-Fi (2.4 GHz)
- Minimum 1 ADC input channel
- 3.3V logic level
- USB interface for programming
- Support for Arduino Framework

Typical examples (or equivalents):

- ESP32 Dev Module
- ESP32 DevKit V1

Selection should prioritize availability, official documentation, and long-term supply stability.

Power Supply Requirements (AC → DC)

The controller and sensors require a stable low-voltage DC power source.

Requirements:

- Input: 110–240V AC (single-phase)
- Output: 5V DC (minimum 1A recommended)
- Electrical isolation between AC input and DC output
- Certified power supply module or adapter

Notes:

- Power supply must comply with local electrical standards
- Use enclosed or DIN-rail mount supplies where applicable

3.2 Sensing Components

Current Sensor Type and Form Factor

The system uses **current-based sensing** to determine device ON/OFF status.

Preferred sensor type:

- Non-invasive AC current sensor (clamp-type)

Form factor:

- Split-core or clamp sensor allowing installation without disconnecting power cables

Typical examples (or equivalents):

- SCT-013 series
- Hall-effect based current sensor modules

Measurement Range Requirements

- Suitable for low to moderate current loads (e.g., computers, UPS, generators)
- Typical range: 0–30A AC (or higher if required by device)
- Accuracy requirements are minimal; detection reliability is the priority

Electrical Isolation Requirements

- Sensor must provide galvanic isolation between AC mains and output signal
- No direct electrical connection between sensor output and high-voltage conductors

This requirement is mandatory for safety and system reliability.

3.3 Supporting Components

Signal Conditioning Components

Depending on the sensor type, the following may be required:

- Burden resistors
- Voltage divider networks
- Simple RC filtering components

These components ensure sensor output is compatible with ESP32 ADC input levels.

Protection Components

To protect the controller and sensors:

- Inline fuses on AC input (where applicable)

- Current-limiting resistors on signal lines
- Transient voltage protection if operating in electrically noisy environments

Enclosure Requirements

- Non-conductive enclosure (plastic or insulated material)
- Adequate ventilation for heat dissipation
- Protection against accidental contact with electrical components
- Wall-mount or DIN-rail mount preferred for structured installations

Connectors and Wiring

- Insulated terminal blocks for AC wiring
- Standard jumper wires or locking connectors for low-voltage signals
- Appropriately rated cables for AC current levels

All wiring must comply with local electrical installation standards.

3.4 Optional / Alternative Components

Smart Plug (Commercial Off-the-Shelf Option)

As an alternative to custom hardware:

- Wi-Fi enabled smart plugs with power monitoring
- Built-in current and voltage measurement
- Native or MQTT-based integration

Examples:

- Sonoff POW series
- Shelly Plug series

This option reduces installation effort but may limit customization.

Voltage Sensor Alternative (If Required)

If voltage presence detection is specifically required:

- Isolated AC voltage sensor modules (e.g., ZMPT101B)

Note:

Voltage-based sensing introduces higher electrical risk and should only be used when current sensing is not feasible.

Industrial-Grade Sensor Options

For harsh or industrial environments:

- DIN-rail mounted current transducers
- Industrial CTs with certified isolation
- Higher temperature and EMI tolerance

3.5 Compliance & Safety Notes

Electrical Safety Considerations

- No direct exposure of low-voltage electronics to AC mains
- All AC connections must be insulated and strain-relieved
- Installation to be performed by qualified personnel only

Installation Constraints

- Sensors must be installed on accessible power lines
- Adequate space required for enclosure and wiring
- Wi-Fi signal availability at installation site

Environmental Considerations

- Indoor or semi-industrial use
- Avoid moisture, excessive dust, and extreme temperatures
- Enclosure selection must match deployment environment